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Study Material Mathematics-I (MA101)

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Mathematics-I (MA101)

Unit-III

Vector Differential Calculus:

Derivatives of vector valued functions, vector equations of curves, tangents of a curve, gradient, directional derivative, divergence and curl, line and double integrals.

1. APPLICATION OF VECTOR CALCULUS

The application of vector calculus to geometry is a field known as **differential geometry**.

Bodies that move in space form paths that may be represented by curves C . This and other applications show the need for **parametric representations** of C with **parameter** t , which may denote time or something else (see Fig.).

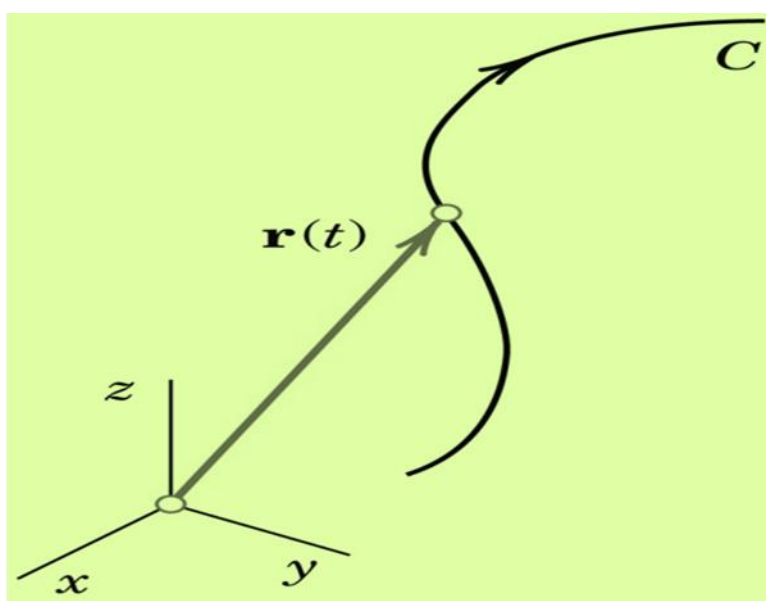


Fig. Parametric representation of a curve

Here t is the parameter and x, y, z are Cartesian coordinates, that is, the usual rectangular coordinates.

A typical parametric representation is given by

$$(1) \quad \vec{r}(t) = [x(t), y(t), z(t)] = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}.$$

2. CURVES, ARC LENGTH, CURVATURE, TORSION:

To each value $t = t_0$, there corresponds a point of C with position vector $\vec{r}(t_0)$ whose coordinates are $x(t_0), y(t_0), z(t_0)$.

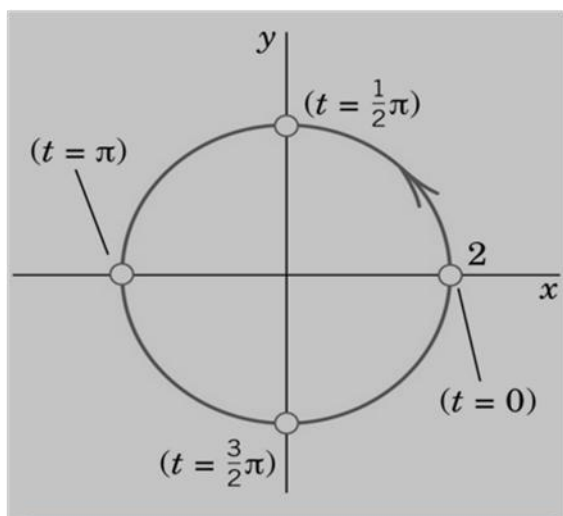


Fig. Circle in Example 1

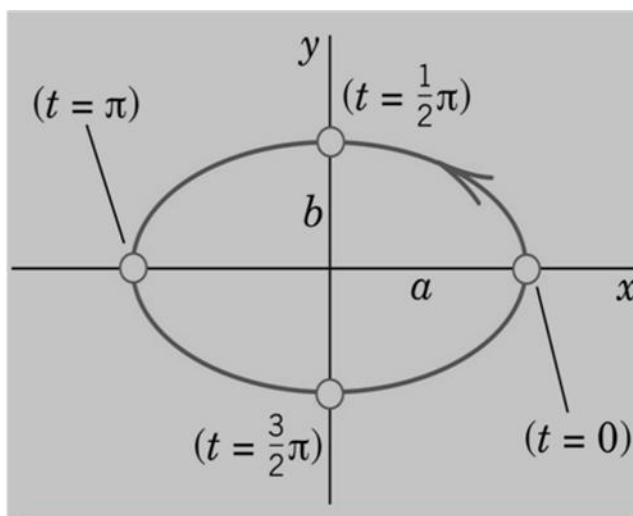


Fig. Ellipse in Example 2

The use of parametric representations has key advantages over other representations that involve projections into the xy -plane and xz -plane or involve a pair of equations with y or with z as independent variable. The projections have the representation

$$(2) \quad y = f(x), \quad z = g(x).$$

The advantages of using (1) instead of (2) are that, in (1), the coordinates x, y, z all play an equal role, that is, all three coordinates are independent variables. Moreover, the parametric representation (1) induces an orientation on C . This means that as we increase t , we travel along the curve C in a certain direction. The sense of increasing t is called the positive sense on C . The sense of decreasing t is then called the negative sense on C , given by (1).

2.1 PARAMETRIC REPRESENTATION OF STRAIGHT LINE:

We know that a straight line passing through a point, say (x_1, y_1, z_1) and having direction ratios, say $l, m,$ and n , can be represented as:

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n}$$

(3)

A *plane curve* is a curve that lies in a plane in space. A curve that is not plane is called a *twisted curve*.

A *simple curve* is a curve without *multiple points*, that is, without points at which the curve intersects or touches itself. Circle and helix are simple curves. Figure given below shows curves that are not simple.

An *arc* of a curve is the portion between any two points of the curve. For simplicity, we say “curve” for curves as well as for arcs.

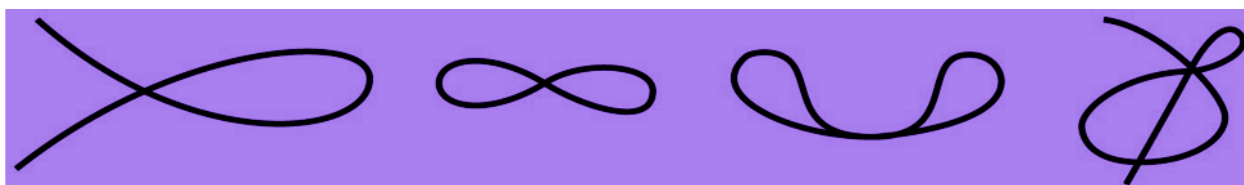


Fig. Curves with multiple points

Tangents are straight lines touching a curve. The tangent to a simple curve C at a point P is the limiting position of a straight line L through P and a point Q of C as Q approaches P along C . If C is given by $\vec{r}(t)$, and P and Q correspond to t and $t + \Delta t$, then a vector in the direction of L is

$$(4) \quad \frac{1}{\Delta t} [\vec{r}(t + \Delta t) - \vec{r}(t)].$$

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Unit Vector: (5) $\vec{u} = \frac{1}{|\vec{r}'|} \vec{r}'$

Tangent to a Curve:

The *tangent* to C at P is given by

(6) $\vec{q}(w) = \vec{r} + w\vec{r}'$

This is the sum of the position vector $\mathbf{r}$ of P and a multiple of the tangent vector $\mathbf{r}'$ of C at P . Both vectors depend on P . The variable w is the parameter in (6).

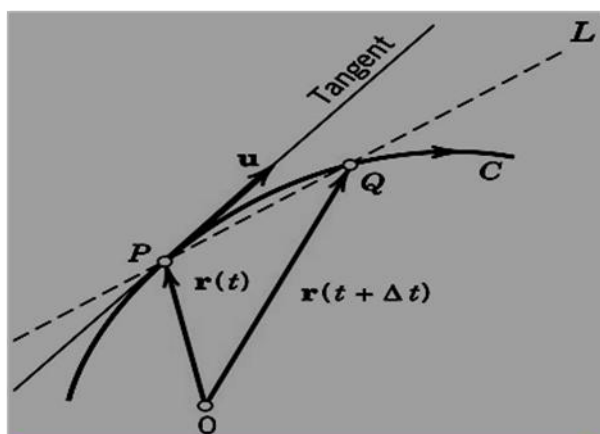


Fig. Tangent to a curve

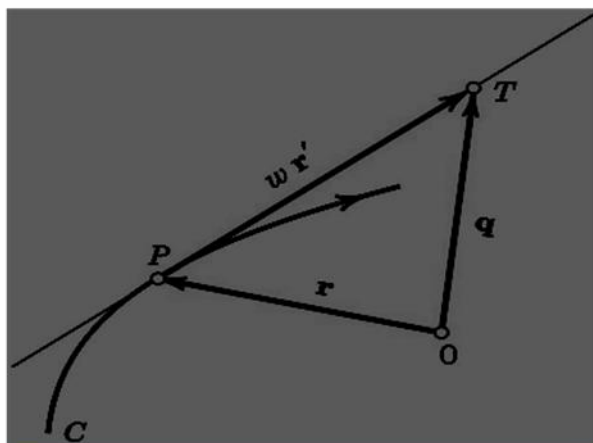


Fig. Formula (6) for the tangent to a curve



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2.2 LENGTH OF A CURVE:

If (t) has a continuous derivative, it can be shown that the sequence $l_1, l_2, \dots$ has a limit, which is independent of the particular choice of the representation of C and of the choice of subdivisions. This limit is given by the integral

$$(7) \quad l = \int_a^b \sqrt{\vec{r}' \bullet \vec{r}'} dt$$

l is called the length of C , and C is called rectifiable.

PROBLEM:

If s is arc length of a curve $C : \vec{r}(t)$

then prove that $\frac{d\vec{r}}{ds}$ is a unit tangent of C .

SOLUTION:

We have $s(t) = \int_a^t \sqrt{\vec{r}' \bullet \vec{r}'} d\tilde{t} \quad \left(\vec{r}' = \frac{d\vec{r}}{d\tilde{t}} \right).$

$$\Rightarrow \frac{ds}{dt} = \sqrt{\frac{d\vec{r}}{dt} \bullet \frac{d\vec{r}}{dt}} = \left| \frac{d\vec{r}}{dt} \right| \text{---(A)}$$

(By Fundamental Theorem of Calculus)

By Chain Rule we have

$$\frac{d\vec{r}}{dt} = \frac{d\vec{r}}{ds} \frac{ds}{dt}$$



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Therefore, $\frac{d\vec{r}}{dt} \bullet \frac{d\vec{r}}{dt} = \left(\frac{d\vec{r}}{ds} \frac{ds}{dt} \right) \bullet \left(\frac{d\vec{r}}{ds} \frac{ds}{dt} \right)$

$$\Rightarrow \left| \frac{d\vec{r}}{dt} \right|^2 = \left(\frac{d\vec{r}}{ds} \bullet \frac{d\vec{r}}{ds} \right) \left(\frac{ds}{dt} \right)^2$$

$$\Rightarrow \left| \frac{d\vec{r}}{dt} \right|^2 = \left| \frac{d\vec{r}}{ds} \right|^2 \left(\frac{ds}{dt} \right)^2$$

$$\Rightarrow \left(\frac{ds}{dt} \right)^2 = \left| \frac{d\vec{r}}{ds} \right|^2 \left(\frac{ds}{dt} \right)^2 \quad \text{Using (A)}$$

$$\Rightarrow \left| \frac{d\vec{r}}{ds} \right|^2 = 1 \quad \text{or} \quad \left| \frac{d\vec{r}}{ds} \right| = 1$$

Moreover, since $\frac{d\vec{r}}{dt} = \frac{d\vec{r}}{ds} \frac{ds}{dt}$,

$\frac{d\vec{r}}{ds}$ is parallel to $\frac{d\vec{r}}{dt}$

Therefore, $\frac{d\vec{r}}{ds}$ is a unit vector which is a tangent of C ,

i.e. $\frac{d\vec{r}}{ds}$ is a unit tangent of C .



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3. DIRECTIONAL DERIVATIVES AND GRADIENT OF A SCALAR FIELD:

3.1 GRADIENT OF A SCALAR FIELD

Let $\vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$ a differential operator.

The gradient of a scalar function f defined as

$$\text{grad } f = \vec{\nabla} f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

EXAMPLE: Find the gradient of $f(x,y,z)=xyz$ at the point $P(1,2,3)$.

Solution: Given function is $f(x,y,z) = xyz$

$$\begin{aligned} \text{By definition, } \text{grad } f &= \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k} \\ &= \frac{\partial}{\partial x} (xyz) \hat{i} + \frac{\partial}{\partial y} (xyz) \hat{j} + \frac{\partial}{\partial z} (xyz) \hat{k} \\ &= yz \hat{i} + xz \hat{j} + xy \hat{k} \end{aligned}$$

DIRECTIONAL DERIVATIVES:

Let $f(x, y, z)$ be any scalar function. Then its *directional derivatives* at the point P in the direction of a vector is the rate of change of f at P in the direction of , which is denoted by

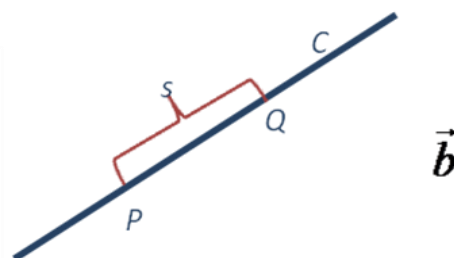
$$D_{\vec{b}} f = \frac{df}{ds} = \lim_{s \rightarrow 0} \frac{f(Q) - f(P)}{s}$$

[s is the distance between P and Q]

Where Q is a variable point on any ray C in the direction of

If $\vec{p}_0$ is the position vector of P and $\vec{b}$ is the unit vector, then the ray C is given by

$$\vec{r}(s) = x(s)\hat{i} + y(s)\hat{j} + z(s)\hat{k} = \vec{p}_0 + s\vec{b} \quad \text{---(1)}$$



In the Cartesian system f is a function of the Cartesian co-ordinates x, y, z , say $f(x, y, z)$, and x, y, z depend on s

By Chain Rule,

$$D_{\vec{b}} f = \frac{df}{ds} = \frac{\partial f}{\partial x} \frac{dx}{ds} + \frac{\partial f}{\partial y} \frac{dy}{ds} + \frac{\partial f}{\partial z} \frac{dz}{ds}$$

$$= \left(\frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k} \right) \cdot \left(\frac{dx}{ds} \hat{i} + \frac{dy}{ds} \hat{j} + \frac{dz}{ds} \hat{k} \right)$$

$$D_{\vec{b}} f = (\text{grad } f) \cdot \frac{d\vec{r}}{ds} \Rightarrow D_{\vec{b}} f = \frac{d\vec{r}}{ds} \cdot (\text{grad } f) \quad \text{---(2)}$$



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$$(1) \Rightarrow \frac{d\vec{r}}{ds} = \frac{dx}{ds} \hat{i} + \frac{dy}{ds} \hat{j} + \frac{dz}{ds} \hat{k} = \frac{d}{ds} (\vec{p}_0 + s\vec{b}) = \vec{b} \quad \text{---(3)}$$

So, from (2) and (3), we get $D_{\vec{b}} f = \vec{b} \cdot (\text{grad } f) \quad \text{---(4)}$

In general the directional derivative of the function f in the direction of a vector $\vec{a} \neq \vec{0}$ of any length is given by

$$D_{\vec{a}} f = \frac{\vec{a}}{|\vec{a}|} \cdot (\text{grad } f) \quad \text{or} \quad D_{\vec{a}} f = \frac{1}{|\vec{a}|} (\vec{a} \cdot \text{grad } f) \quad \text{---(5)}$$

SOME RESULTS CONCERNING GRADIENT

THEOREM: Let f be a differentiable scalar function that represents a surface $S: f(x, y, z) = c = \text{constant}$. Then if the gradient of f at a point P of S is not the zero vector, it is a normal vector of S at P .

Solution:



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Let $f(x,y,z)=c$ -----(1) represents a surface, say S .

$$\text{Let } \vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k} \text{ --- (2)}$$

be the position vector any curve, say C on S .

$$\text{Now, } \frac{df}{dt} = \frac{d}{dt}(c) = 0$$

$$\Rightarrow \frac{d}{dt}[f(x(t), y(t), z(t))] = 0$$

$$\Rightarrow \left(\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} \right) = 0$$

$$\Rightarrow \left(\frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k} \right) \cdot \left(\frac{dx}{dt} \hat{i} + \frac{dy}{dt} \hat{j} + \frac{dz}{dt} \hat{k} \right) = 0$$

$$\Rightarrow \vec{\nabla} f \cdot \frac{d\vec{r}}{dt} = 0 \quad \Rightarrow (\text{grad } f) \cdot \vec{r}' = 0$$

As $\vec{r}'$ represents the tangent vector to the surface and $(\text{grad } f) \cdot \vec{r}' = 0$

$\Rightarrow \text{grad } f$ is perpendicular to the tangent vector at the point P .

$\Rightarrow \text{grad } f$ is normal to the surface S at P .

SOME USEFUL RESULTS

$$1. \vec{\nabla}(fg) = f\vec{\nabla}g + g\vec{\nabla}f$$



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Proof:

Proof: $\vec{\nabla}(fg) = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (fg)$

$$= \frac{\partial}{\partial x} (fg) \hat{i} + \frac{\partial}{\partial y} (fg) \hat{j} + \frac{\partial}{\partial z} (fg) \hat{k}$$

$$= \left(f \frac{\partial g}{\partial x} + g \frac{\partial f}{\partial x} \right) \hat{i} + \left(f \frac{\partial g}{\partial y} + g \frac{\partial f}{\partial y} \right) \hat{j} + \left(f \frac{\partial g}{\partial z} + g \frac{\partial f}{\partial z} \right) \hat{k}$$

$$= f \left(\frac{\partial g}{\partial x} \hat{i} + \frac{\partial g}{\partial y} \hat{j} + \frac{\partial g}{\partial z} \hat{k} \right) + g \left(\frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k} \right) = f \vec{\nabla} g + g \vec{\nabla} f$$

2. Prove that:

$$\nabla^2(fg) = g \nabla^2 f + f \nabla^2 g + 2 \vec{\nabla} f \cdot \vec{\nabla} g$$

Proof:

$$\begin{aligned} \nabla^2(fg) &= \frac{\partial^2}{\partial x^2}(fg) + \frac{\partial^2}{\partial y^2}(fg) + \frac{\partial^2}{\partial z^2}(fg) \\ &= \frac{\partial}{\partial x} \left(\frac{\partial}{\partial x}(fg) \right) + \frac{\partial}{\partial y} \left(\frac{\partial}{\partial y}(fg) \right) + \frac{\partial}{\partial z} \left(\frac{\partial}{\partial z}(fg) \right) \\ &= \frac{\partial}{\partial x} \left(f \frac{\partial g}{\partial x} + g \frac{\partial f}{\partial x} \right) + \frac{\partial}{\partial y} \left(f \frac{\partial g}{\partial y} + g \frac{\partial f}{\partial y} \right) + \frac{\partial}{\partial z} \left(f \frac{\partial g}{\partial z} + g \frac{\partial f}{\partial z} \right) \\ &= f \frac{\partial^2 g}{\partial x^2} + \frac{\partial g}{\partial x} \frac{\partial f}{\partial x} + g \frac{\partial^2 f}{\partial x^2} + \frac{\partial f}{\partial x} \frac{\partial g}{\partial x} \\ &\quad + f \frac{\partial^2 g}{\partial y^2} + \frac{\partial g}{\partial y} \frac{\partial f}{\partial y} + g \frac{\partial^2 f}{\partial y^2} + \frac{\partial f}{\partial y} \frac{\partial g}{\partial y} \\ &\quad + f \frac{\partial^2 g}{\partial z^2} + \frac{\partial g}{\partial z} \frac{\partial f}{\partial z} + g \frac{\partial^2 f}{\partial z^2} + \frac{\partial f}{\partial z} \frac{\partial g}{\partial z} \\ &= f \left(\frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2} + \frac{\partial^2 g}{\partial z^2} \right) + g \left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \right) + 2 \left(\frac{\partial f}{\partial x} \frac{\partial g}{\partial x} + \frac{\partial f}{\partial y} \frac{\partial g}{\partial y} + \frac{\partial f}{\partial z} \frac{\partial g}{\partial z} \right) \\ &= f \nabla^2 g + g \nabla^2 f + 2 \vec{\nabla} f \cdot \vec{\nabla} g \end{aligned}$$



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$$\begin{aligned} &= g\nabla^2 f + f\nabla^2 g + 2\left(\frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k}\right) \cdot \left(\frac{\partial g}{\partial x}\hat{i} + \frac{\partial g}{\partial y}\hat{j} + \frac{\partial g}{\partial z}\hat{k}\right) \\ &= g\nabla^2 f + f\nabla^2 g + 2\vec{\nabla}f \cdot \vec{\nabla}g \end{aligned}$$

3. Prove that:

$$\vec{\nabla}\left(\frac{f}{g}\right) = \frac{1}{g^2}(g\vec{\nabla}f - f\vec{\nabla}g)$$

Proof:

$$\begin{aligned} \vec{\nabla}\left(\frac{f}{g}\right) &= \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\left(\frac{f}{g}\right) \\ &= \hat{i}\frac{\partial}{\partial x}\left(\frac{f}{g}\right) + \hat{j}\frac{\partial}{\partial y}\left(\frac{f}{g}\right) + \hat{k}\frac{\partial}{\partial z}\left(\frac{f}{g}\right) \\ &= \hat{i}\left(\frac{g\frac{\partial f}{\partial x} - f\frac{\partial g}{\partial x}}{g^2}\right) + \hat{j}\left(\frac{g\frac{\partial f}{\partial y} - f\frac{\partial g}{\partial y}}{g^2}\right) + \hat{k}\left(\frac{g\frac{\partial f}{\partial z} - f\frac{\partial g}{\partial z}}{g^2}\right) \\ &= \frac{1}{g^2}\left[\left(g\left(\frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k}\right) - f\left(\frac{\partial g}{\partial x}\hat{i} + \frac{\partial g}{\partial y}\hat{j} + \frac{\partial g}{\partial z}\hat{k}\right)\right)\right] \\ &= \frac{1}{g^2}(g\vec{\nabla}f - f\vec{\nabla}g) \end{aligned}$$



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Problems:

1. Find the directional derivative of $f(x,y)=e^x \cos y$ at the point $p(2,\pi,0)$ in the direction of the vector $\vec{a} = 2\hat{i} + 3\hat{j}$.

Solution:

Given function is $f(x,y,z) = e^x \cos y$

$$\begin{aligned}\therefore \text{grad } f &= \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k} \\ &= \frac{\partial}{\partial x} (e^x \cos y) \hat{i} + \frac{\partial}{\partial y} (e^x \cos y) \hat{j} + \frac{\partial}{\partial z} (e^x \cos y) \hat{k} \\ &= e^x \cos y \hat{i} - e^x \sin y \hat{j}\end{aligned}$$

$$\text{Now } [\nabla f]_{(2,\pi,0)} = (e^2 \cos \pi) \hat{i} + (e^2 \sin \pi) \hat{j} = -e^2 \hat{i}$$

$$\begin{aligned}D_{\vec{a}} f &= \frac{\vec{a}}{|\vec{a}|} \cdot \nabla f \\ &= \left(\frac{2\hat{i} + 3\hat{j}}{|2\hat{i} + 3\hat{j}|} \right) \cdot (-e^2 \hat{i}) \\ &= \left(\frac{2\hat{i} + 3\hat{j}}{\sqrt{13}} \right) \cdot (-e^2 \hat{i}) = -\frac{2}{\sqrt{13}} e^2\end{aligned}$$



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ASSIGNMENTS

1. Find the directional derivative of $f(x, y, z) = \frac{1}{x^2 + y^2 + z^2}$

at $P(3, 0, 4)$ in the direction of $\vec{a} = \hat{i} + \hat{j} + \hat{k}$.

2. Find f if $\nabla f = [ye^x \quad e^x \quad 1]$

3. Prove that $\nabla^2(fg) = g\nabla^2 f + f\nabla^2 g + 2\nabla f \cdot \nabla g$

4. Find the unit normal vector to the surface

$$x^2 + y^2 + 2z^2 = 26 \text{ at } P(2, 2, 3).$$

4. DIVERGENCE OF A VECTOR FIELD:

The divergence of a vector function $\vec{v}(x, y, z) = v_1 \hat{i} + v_2 \hat{j} + v_3 \hat{k}$

is a scalar function which is defined as

$$\text{div } \vec{v} = \vec{\nabla} \cdot \vec{v}$$

$$= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (v_1 \hat{i} + v_2 \hat{j} + v_3 \hat{k})$$

$$= \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$



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4.1 EXAMPLE OF DIVERGENCE:

Example: Find the divergence of $2x^2z\hat{i} - xy^2\hat{j} + 3yz^2\hat{k}$.

Solution:

$$\text{Let } \vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k} = 2x^2z\hat{i} - xy^2\hat{j} + 3yz^2\hat{k}$$

$$\begin{aligned} \text{div } \vec{v} &= \left(\frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z} \right) \\ &= \frac{\partial}{\partial x}(2x^2z) + \frac{\partial}{\partial y}(-xy^2) + \frac{\partial}{\partial z}(3yz^2) \\ &= 4xz - 2xy + 6yz \end{aligned}$$

4.2 PHYSICAL SIGNIFICANCE OF DIVERGENCE

The divergence of a vector function at a point P constitute the volume density of the flux of the vector function at that point that is divergence measures the change of flow. Now if the density is constant the fluid is incompressible and for that case

$$\text{div } \vec{v} = 0$$

4.3 USEFUL RESULTS INVOLVING DIVERGENCE

1. Prove that: $\text{div}(k\vec{v}) = k \text{div } \vec{v}$

Proof: Let $\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$

$$\therefore k\vec{v} = k(v_1\hat{i} + v_2\hat{j} + v_3\hat{k}) = (kv_1)\hat{i} + (kv_2)\hat{j} + (kv_3)\hat{k}$$

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$$\therefore \text{div}(k\vec{v}) = \vec{\nabla} \cdot (k\vec{v}) = \left(\frac{\partial}{\partial x}(kv_1) + \frac{\partial}{\partial y}(kv_2) + \frac{\partial}{\partial z}(kv_3) \right)$$



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2. Prove that $\text{div}(f\vec{v}) = f \text{div} \vec{v} + \vec{v} \cdot \vec{\nabla} f$

Proof:

$$\text{Let } \vec{v} = v_1 \hat{i} + v_2 \hat{j} + v_3 \hat{k}$$

$$\therefore f\vec{v} = f(v_1 \hat{i} + v_2 \hat{j} + v_3 \hat{k}) = (fv_1) \hat{i} + (fv_2) \hat{j} + (fv_3) \hat{k}$$

$$\therefore \text{div}(f\vec{v}) = \vec{\nabla} \cdot (f\vec{v}) = \frac{\partial}{\partial x}(fv_1) + \frac{\partial}{\partial y}(fv_2) + \frac{\partial}{\partial z}(fv_3)$$

$$= \left(f \frac{\partial v_1}{\partial x} + v_1 \frac{\partial f}{\partial x} \right) + \left(f \frac{\partial v_2}{\partial y} + v_2 \frac{\partial f}{\partial y} \right) + \left(f \frac{\partial v_3}{\partial z} + v_3 \frac{\partial f}{\partial z} \right)$$

$$= f \left(\frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z} \right) + \left(v_1 \frac{\partial f}{\partial x} + v_2 \frac{\partial f}{\partial y} + v_3 \frac{\partial f}{\partial z} \right)$$

$$= f \text{div} \vec{v} + (v_1 \hat{i} + v_2 \hat{j} + v_3 \hat{k}) \cdot \left(\frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k} \right)$$

$$= f \text{div} \vec{v} + \vec{v} \cdot \vec{\nabla} f$$



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3. Prove that $\text{div}(f \vec{\nabla} g) = f \nabla^2 g + (\vec{\nabla} f \cdot \vec{\nabla} g)$

Proof:

$$\vec{\nabla} g = \frac{\partial g}{\partial x} \hat{i} + \frac{\partial g}{\partial y} \hat{j} + \frac{\partial g}{\partial z} \hat{k}$$

$$\begin{aligned} \therefore \text{div}(f \vec{\nabla} g) &= \vec{\nabla} \cdot (f \vec{\nabla} g) = \frac{\partial}{\partial x} \left(f \frac{\partial g}{\partial x} \right) + \frac{\partial}{\partial y} \left(f \frac{\partial g}{\partial y} \right) + \frac{\partial}{\partial z} \left(f \frac{\partial g}{\partial z} \right) \\ &= \left(f \frac{\partial^2 g}{\partial x^2} + \frac{\partial g}{\partial x} \frac{\partial f}{\partial x} \right) + \left(f \frac{\partial^2 g}{\partial y^2} + \frac{\partial g}{\partial y} \frac{\partial f}{\partial y} \right) + \left(f \frac{\partial^2 g}{\partial z^2} + \frac{\partial g}{\partial z} \frac{\partial f}{\partial z} \right) \\ &= f \left(\frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2} + \frac{\partial^2 g}{\partial z^2} \right) + \left(\frac{\partial g}{\partial x} \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} \frac{\partial f}{\partial y} + \frac{\partial g}{\partial z} \frac{\partial f}{\partial z} \right) \\ &= f \nabla^2 g + \left(\frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k} \right) \cdot \left(\frac{\partial g}{\partial x} \hat{i} + \frac{\partial g}{\partial y} \hat{j} + \frac{\partial g}{\partial z} \hat{k} \right) \\ &= f \nabla^2 g + (\vec{\nabla} f \cdot \vec{\nabla} g) \end{aligned}$$

PRACTICE PROBLEMS:

1. Find the value of $\nabla^2 f$ if $f = \frac{xy}{z}$.

Solution: Given function is $f = \frac{xy}{z}$

$$\therefore \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} \left(\frac{xy}{z} \right) = \frac{y}{z}, \quad \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} \left(\frac{xy}{z} \right) = \frac{x}{z} \quad \text{and} \quad \frac{\partial f}{\partial z} = \frac{\partial}{\partial z} \left(\frac{xy}{z} \right) = -\frac{xy}{z^2}$$

$$\Rightarrow \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{y}{z} \right) = 0, \quad \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{x}{z} \right) = 0, \quad \frac{\partial^2 f}{\partial z^2} = \frac{\partial}{\partial z} \left(-\frac{xy}{z^2} \right) = \frac{2xy}{z^3}$$

$$\text{So} \quad \nabla^2 f = \left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \right) = 0 + 0 + \frac{2xy}{z^3} = \frac{2xy}{z^3}$$



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2. Find the divergence of $e^{-x}\hat{i} + e^{-x}y\hat{j} + 2z\sinh x\hat{k}$.

Solution:

$$\begin{aligned} & \operatorname{div}\left(e^{-x}\hat{i} + e^{-x}y\hat{j} + 2z\sinh x\hat{k}\right) \\ &= \frac{\partial}{\partial x}(e^{-x}) + \frac{\partial}{\partial y}(e^{-x}y) + \frac{\partial}{\partial z}(2z\sinh x) \\ &= e^{-x} + e^{-x} + 2\sinh x \\ &= e^{-x} + e^{-x} + e^x - e^{-x} \\ &= 2e^{-x} \end{aligned}$$

6. CURL OF A VECTOR FIELD

6.1 DEFINITION OF CURL

The *curl operator* denoted by $\nabla \times$, acts on a vector and gives another vector as a result.

If $\mathbf{A} = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}$, then $\operatorname{curl} \mathbf{A} = \nabla \times \mathbf{A}$.

$$\text{i.e. } \operatorname{curl} \mathbf{A} = \nabla \times \mathbf{A} = \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}\right) \times (a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k})$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_x & a_y & a_z \end{vmatrix}$$

$$\therefore \nabla \times \mathbf{A} = \mathbf{i} \left(\frac{\partial a_z}{\partial y} - \frac{\partial a_y}{\partial z}\right) + \mathbf{j} \left(\frac{\partial a_x}{\partial z} - \frac{\partial a_z}{\partial x}\right) + \mathbf{k} \left(\frac{\partial a_y}{\partial x} - \frac{\partial a_x}{\partial y}\right)$$

$\operatorname{Curl} \mathbf{A}$ is thus a vector function. It is best remembered in its determinant form, so make a note of it.

If $\nabla \times \mathbf{A} = \mathbf{0}$ then $\mathbf{A}$ is said to be *irrotational*.



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Example:

If $\mathbf{A} = (y^4 - x^2z^2)\mathbf{i} + (x^2 + y^2)\mathbf{j} - x^2yz\mathbf{k}$, determine $\text{curl } \mathbf{A}$ at the point $(1, 3, -2)$.

$$\text{curl } \mathbf{A} = \nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^4 - x^2z^2 & x^2 + y^2 & -x^2yz \end{vmatrix}$$

Now we expand the determinant

$$\begin{aligned} \nabla \times \mathbf{A} = & \mathbf{i} \left\{ \frac{\partial}{\partial y} (-x^2yz) - \frac{\partial}{\partial z} (x^2 + y^2) \right\} - \mathbf{j} \left\{ \frac{\partial}{\partial x} (-x^2yz) - \frac{\partial}{\partial z} (y^4 - x^2z^2) \right\} \\ & + \mathbf{k} \left\{ \frac{\partial}{\partial x} (x^2 + y^2) - \frac{\partial}{\partial y} (y^4 - x^2z^2) \right\} \end{aligned}$$

All that now remains is to obtain the partial derivatives and substitute the values of x, y, z .

$$\nabla \times \mathbf{A} = \mathbf{i}\{-x^2z\} - \mathbf{j}\{-2xyz + 2x^2z\} + \mathbf{k}\{2x - 4y^3\}.$$

$$\begin{aligned} \therefore \text{At } (1, 3, -2), \quad \nabla \times \mathbf{A} &= \mathbf{i}(2) - \mathbf{j}(12 - 4) + \mathbf{k}(2 - 108) \\ &= 2\mathbf{i} - 8\mathbf{j} - 106\mathbf{k} \end{aligned}$$

6.2 PHYSICAL SIGNIFICANCE OF CURL

The curl of a vector function at a point P gives the angular velocity at that point of the vector field. A vector function is said to be **irrational** if

$$\text{curl } \vec{v} = \vec{0}$$

at every point otherwise it is rotational.

6.3 SOME USEFUL RESULTS ON CURL

1. Prove that $\text{curl}(\vec{u} + \vec{v}) = \text{curl } \vec{u} + \text{curl } \vec{v}$

Proof: Let $\vec{u} = u_1\hat{i} + u_2\hat{j} + u_3\hat{k}$ and $\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$

$$\begin{aligned} \therefore \vec{u} + \vec{v} &= (u_1\hat{i} + u_2\hat{j} + u_3\hat{k}) + (v_1\hat{i} + v_2\hat{j} + v_3\hat{k}) \\ &= (u_1 + v_1)\hat{i} + (u_2 + v_2)\hat{j} + (u_3 + v_3)\hat{k} \end{aligned}$$

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$$\begin{aligned} \text{curl}(\vec{u} + \vec{v}) = \vec{\nabla} \times (\vec{u} + \vec{v}) &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u_1 + v_1 & u_2 + v_2 & u_3 + v_3 \end{vmatrix} \\ &= \left[\frac{\partial}{\partial y}(u_3 + v_3) - \frac{\partial}{\partial z}(u_2 + v_2) \right] \hat{i} \\ &\quad + \left[\frac{\partial}{\partial z}(u_1 + v_1) - \frac{\partial}{\partial x}(u_3 + v_3) \right] \hat{j} \\ &\quad + \left[\frac{\partial}{\partial x}(u_2 + v_2) - \frac{\partial}{\partial y}(u_1 + v_1) \right] \hat{k} \end{aligned}$$

$$\begin{aligned} &= \left[\left(\frac{\partial u_3}{\partial y} + \frac{\partial v_3}{\partial y} \right) - \left(\frac{\partial u_2}{\partial z} + \frac{\partial v_2}{\partial z} \right) \right] \hat{i} \\ &\quad + \left[\left(\frac{\partial u_1}{\partial z} + \frac{\partial v_1}{\partial z} \right) - \left(\frac{\partial u_3}{\partial x} + \frac{\partial v_3}{\partial x} \right) \right] \hat{j} \\ &\quad + \left[\left(\frac{\partial u_2}{\partial x} + \frac{\partial v_2}{\partial x} \right) - \left(\frac{\partial u_1}{\partial y} + \frac{\partial v_1}{\partial y} \right) \right] \hat{k} \\ &= \left[\left(\frac{\partial u_3}{\partial y} - \frac{\partial u_2}{\partial z} \right) \hat{i} + \left(\frac{\partial u_1}{\partial z} - \frac{\partial u_3}{\partial x} \right) \hat{j} + \left(\frac{\partial u_2}{\partial x} - \frac{\partial u_1}{\partial y} \right) \hat{k} \right] \\ &\quad + \left[\left(\frac{\partial v_3}{\partial y} - \frac{\partial v_2}{\partial z} \right) \hat{i} + \left(\frac{\partial v_1}{\partial z} - \frac{\partial v_3}{\partial x} \right) \hat{j} + \left(\frac{\partial v_2}{\partial x} - \frac{\partial v_1}{\partial y} \right) \hat{k} \right] \end{aligned}$$



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2. $\text{div}(\text{curl } \vec{v}) = 0$

Solution: Let $\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$

$$\begin{aligned}\text{By definition, } \text{curl } \vec{v} &= \vec{\nabla} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix} = \\ &= \left(\frac{\partial v_3}{\partial y} - \frac{\partial v_2}{\partial z} \right) \hat{i} + \left(\frac{\partial v_1}{\partial z} - \frac{\partial v_3}{\partial x} \right) \hat{j} + \left(\frac{\partial v_2}{\partial x} - \frac{\partial v_1}{\partial y} \right) \hat{k} \\ \text{div}(\text{curl } \vec{v}) &= \vec{\nabla} \cdot (\text{curl } \vec{v}) = \frac{\partial}{\partial x} \left(\frac{\partial v_3}{\partial y} - \frac{\partial v_2}{\partial z} \right) \\ &+ \frac{\partial}{\partial y} \left(\frac{\partial v_1}{\partial z} - \frac{\partial v_3}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial v_2}{\partial x} - \frac{\partial v_1}{\partial y} \right) \\ &= \frac{\partial^2 v_3}{\partial x \partial y} - \frac{\partial^2 v_2}{\partial x \partial z} + \frac{\partial^2 v_1}{\partial y \partial z} - \frac{\partial^2 v_3}{\partial y \partial x} + \frac{\partial^2 v_2}{\partial z \partial x} - \frac{\partial^2 v_1}{\partial z \partial y} \\ &= \frac{\partial^2 v_3}{\partial x \partial y} - \frac{\partial^2 v_2}{\partial z \partial x} + \frac{\partial^2 v_1}{\partial y \partial z} - \frac{\partial^2 v_3}{\partial x \partial y} + \frac{\partial^2 v_2}{\partial z \partial x} - \frac{\partial^2 v_1}{\partial y \partial z} \\ &= 0\end{aligned}$$

ASSIGNMENTS

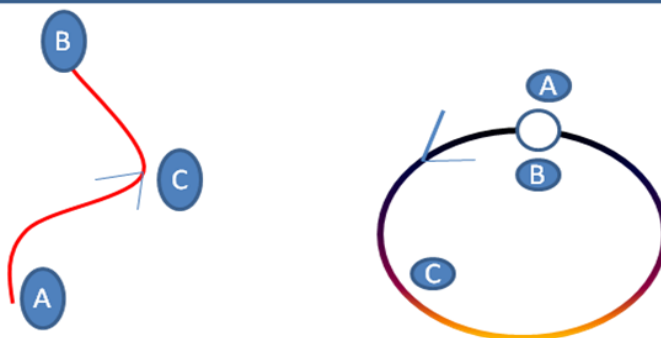
1. Find the curl of $\vec{v} = yz\hat{i} + 3zx\hat{j} + z\hat{k}$ at $P(2,3,4)$
2. Show that $\text{div}(\text{curl } \vec{v}) = 0$.

7. LINE INTEGRALS

7.1 INTRODUCTION TO LINE INTEGRAL

Let C be a curve with a parametric representation

$$\vec{r}(t) = [x(t), y(t), z(t)] = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$



The curve C is called path of integration, $A: \vec{r}(a)$ its initial point and $B: \vec{r}(b)$ its terminal point of the path of integration C . The direction from A to B in which t increases is called the positive direction on C . We can indicate the direction by arrow. If the point A and B coincide, then C is called a closed path.

7.2 Physical Applications of Line Integrals

In physics, the line integrals are used, in particular, for computations of

- mass of a wire;
- center of mass and moments of inertia of a wire;
- work done by a force on an object moving in a vector field;
- magnetic field around a conductor (Ampere's Law);
- voltage generated in a loop (Faraday's Law of magnetic induction).

7.3 General Properties of Line Integral:

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$$1. \int_C k\vec{F}(\vec{r}).d\vec{r} = k \int_C \vec{F}(\vec{r}).d\vec{r} \text{ where } k \text{ is a constant}$$

2. For vector functions $\vec{G}_1, \vec{G}_2, \vec{G}_3, \dots, \vec{G}_n$ whose line integral of the form (1) exists and $a_1, a_2, a_3, \dots, a_n$ arbitrary constants

$$\begin{aligned} & \int_C \left(a_1 \vec{G}_1(\vec{r}) + a_2 \vec{G}_2(\vec{r}) + a_3 \vec{G}_3(\vec{r}) + \dots + a_n \vec{G}_n(\vec{r}) \right) \cdot d\vec{r} \\ &= a_1 \int_C \vec{G}_1(\vec{r}) \cdot d\vec{r} + a_2 \int_C \vec{G}_2(\vec{r}) \cdot d\vec{r} + a_3 \int_C \vec{G}_3(\vec{r}) \cdot d\vec{r} + \dots + a_n \int_C \vec{G}_n(\vec{r}) \cdot d\vec{r} \end{aligned}$$

$$3. \int_C \vec{F}(\vec{r}).d\vec{r} = \int_{C_1} \vec{F}(\vec{r}).d\vec{r} + \int_{C_2} \vec{F}(\vec{r}).d\vec{r} + \dots + \int_{C_n} \vec{F}(\vec{r}).d\vec{r}$$

where C is the sum of the curve $C_1, C_2, C_3, \dots, C_n$.

$$4. \int_a^b \vec{F}(\vec{r}).d\vec{r} = - \int_b^a \vec{F}(\vec{r}).d\vec{r}$$

7.4 Scalar Field:

If a scalar field V exists for all points on the curve, then

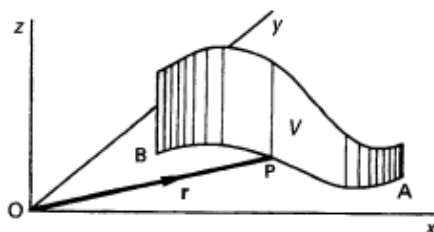
$\sum_{p=1}^n V \mathbf{dr}_p$ with $\mathbf{dr} \rightarrow 0$, defines the *line integral* of V along the curve c from A to B ,

$$\text{i.e. line integral} = \int_c V \mathbf{dr}$$

We can illustrate this integral by erecting a continuous ordinate proportional to V at each point of the curve.

$\int_c V \mathbf{dr}$ is then represented by the area of the curved surface between the ends A and B of the curve c .

To evaluate a line integral, the integrand is expressed in terms of x, y, z , with $\mathbf{dr} = \dots\dots\dots$





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Example:1

If $V = xy + y^2z$, evaluate $\int_c V \, d\mathbf{r}$ along the curve c defined by $x = t^2$; $y = 2t$; $z = t + 5$ between $A (0, 0, 5)$ and $B (4, 4, 7)$.

Solution:

$$V = 6t^3 + 20t^2; \quad d\mathbf{r} = \mathbf{i} 2t \, dt + \mathbf{j} 2 \, dt + \mathbf{k} \, dt$$

Because

$$V = xy + y^2z = (t^2)(2t) + (4t^2)(t + 5) = 6t^3 + 20t^2.$$

$$\text{Also } \left. \begin{array}{l} x = t^2 \quad dx = 2t \, dt \\ y = 2t \quad dy = 2 \, dt \\ z = t + 5 \quad dz = dt \end{array} \right\} \therefore d\mathbf{r} = \mathbf{i} \, dx + \mathbf{j} \, dy + \mathbf{k} \, dz = \mathbf{i} 2t \, dt + \mathbf{j} 2 \, dt + \mathbf{k} \, dt$$

$$\therefore \int_c V \, d\mathbf{r} = \int_c (6t^3 + 20t^2)(\mathbf{i} 2t + \mathbf{j} 2 + \mathbf{k}) \, dt$$

Limits: $A (0, 0, 5) \equiv t = \dots\dots\dots$

$B (4, 4, 7) \equiv t = \dots\dots\dots$

$$A (0, 0, 5) \equiv t = 0; \quad B (4, 4, 7) \equiv t = 2$$

$$\therefore \int_c V \, d\mathbf{r} = \int_0^2 (6t^3 + 20t^2)(\mathbf{i} 2t + \mathbf{j} 2 + \mathbf{k}) \, dt$$

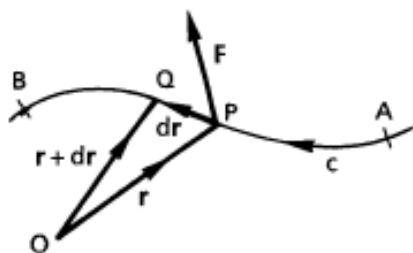
$$\frac{8}{15} (444 \mathbf{i} + 290 \mathbf{j} + 145 \mathbf{k})$$

$$\int_c V \, d\mathbf{r} = 2 \int_0^2 \{(6t^4 + 20t^3)\mathbf{i} + (6t^3 + 20t^2)\mathbf{j} + (3t^3 + 10t^2)\mathbf{k}\} \, dt$$

7.5 Vector Field:

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If a vector field $\mathbf{F}$ exists for all points of the curve c , then for each element of arc we can form the scalar product $\mathbf{F} \cdot d\mathbf{r}$. Summing these products for all elements of arc, we have $\sum_{p=1}^n \mathbf{F} \cdot d\mathbf{r}_p$

Then, if $d\mathbf{r}_p \rightarrow 0$, the sum becomes the integral $\int_c \mathbf{F} \cdot d\mathbf{r}$,
i.e. the line integral of $\mathbf{F}$ from A to B along the stated curve

$$= \int_c \mathbf{F} \cdot d\mathbf{r}$$

In this case, since $\mathbf{F} \cdot d\mathbf{r}$ is a scalar product, then the line integral is a scalar.

To evaluate the line integral, $\mathbf{F}$ and $d\mathbf{r}$ are expressed in terms of x, y, z and the curve in parametric form. We have

$$\mathbf{F} = F_x \mathbf{i} + F_y \mathbf{j} + F_z \mathbf{k}$$

and $d\mathbf{r} = \mathbf{i} dx + \mathbf{j} dy + \mathbf{k} dz$

Then $\mathbf{F} \cdot d\mathbf{r} = (F_x \mathbf{i} + F_y \mathbf{j} + F_z \mathbf{k}) \cdot (\mathbf{i} dx + \mathbf{j} dy + \mathbf{k} dz)$
 $= F_x dx + F_y dy + F_z dz$

$$\therefore \int_c \mathbf{F} \cdot d\mathbf{r} = \int_c F_x dx + \int_c F_y dy + \int_c F_z dz$$

Example:

If $\mathbf{F} = x^2 y \mathbf{i} + xz \mathbf{j} - 2yz \mathbf{k}$, evaluate $\int_c \mathbf{F} \cdot d\mathbf{r}$ between A (0, 0, 0) and B (4, 2, 1) along the curve having parametric equations $x = 4t$; $y = 2t^2$; $z = t^3$.

Solution:



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$$\mathbf{F} = 32t^4 \mathbf{i} + 4t^4 \mathbf{j} - 4t^5 \mathbf{k}$$
$$dx = 4 dt; \quad dy = 4t dt; \quad dz = 3t^2 dt$$

Because

$$\begin{aligned} x^2 y &= (16t^2)(2t^2) = 32t^4 & x &= 4t & \therefore dx &= 4 dt \\ xz &= (4t)(t^3) = 4t^4 & y &= 2t^2 & \therefore dy &= 4t dt \\ 2yz &= (4t^2)(t^3) = 4t^5 & z &= t^3 & \therefore dz &= 3t^2 dt \end{aligned}$$

$$\begin{aligned} \text{Then } \int \mathbf{F} \cdot d\mathbf{r} &= \int (32t^4 \mathbf{i} + 4t^4 \mathbf{j} - 4t^5 \mathbf{k}) \cdot (\mathbf{i} 4 dt + \mathbf{j} 4t dt + \mathbf{k} 3t^2 dt) \\ &= \int (128t^4 + 16t^5 - 12t^7) dt \end{aligned}$$

Limits: A (0, 0, 0) $\equiv t = \dots\dots\dots$; B (4, 2, 1) $\equiv t = \dots\dots\dots$

$$A \equiv t = 0; \quad B \equiv t = 1$$

$$\therefore \int_c \mathbf{F} \cdot d\mathbf{r} = \int_0^1 (128t^4 + 16t^5 - 12t^7) dt = \dots\dots\dots$$

$$\frac{128}{5} + \frac{8}{3} - \frac{3}{2} = \frac{803}{30} = 26.77$$

If the vector field $\mathbf{F}$ is a *force field*, then the line integral $\int_c \mathbf{F} \cdot d\mathbf{r}$ represents the work done in moving a unit particle along the prescribed curve c from A to B.

Example:

If $\mathbf{F} = x^2 y \mathbf{i} + 2yz \mathbf{j} + 3z^2 x \mathbf{k}$, evaluate $\int_c \mathbf{F} \cdot d\mathbf{r}$ between A (0, 0, 0) and B (1, 2, 3)

- (a) along the straight lines c_1 from (0, 0, 0) to (1, 0, 0)
then c_2 from (1, 0, 0) to (1, 2, 0)
and c_3 from (1, 2, 0) to (1, 2, 3)
- (b) along the straight line c_4 joining (0, 0, 0) to (1, 2, 3).

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Solution:

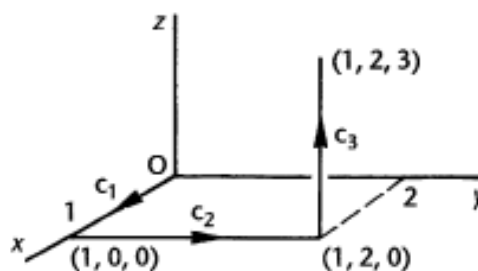
$$\mathbf{F} \cdot d\mathbf{r} = x^2 y dx + 2yz dy + 3z^2 x dz$$

Because

$$\mathbf{F} \cdot d\mathbf{r} = (x^2 y \mathbf{i} + 2yz \mathbf{j} + 3z^2 x \mathbf{k}) \cdot (dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k})$$

$$\therefore \int \mathbf{F} \cdot d\mathbf{r} = \int x^2 y dx + \int 2yz dy + \int 3z^2 x dz$$

(a) Here the integration is made in three sections, along c_1 , c_2 and c_3 .



(1) c_1 : $y = 0, z = 0, dy = 0, dz = 0$

$$\therefore \int_{c_1} \mathbf{F} \cdot d\mathbf{r} = 0 + 0 + 0 = 0$$

(2) c_2 : The conditions along c_2 are

$$c_2: x = 1, z = 0, dx = 0, dz = 0$$

$$\therefore \int_{c_2} \mathbf{F} \cdot d\mathbf{r} = 0 + 0 + 0 = 0$$

(3) c_3 : $x = 1, y = 2, dx = 0, dy = 0$

$$\therefore \int_{c_3} \mathbf{F} \cdot d\mathbf{r} = \dots\dots\dots$$

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Because

$$\int_{C_3} \mathbf{F} \cdot d\mathbf{r} = 0 + 0 + \int_0^3 3z^2 dz = 27$$

Summing the three partial results

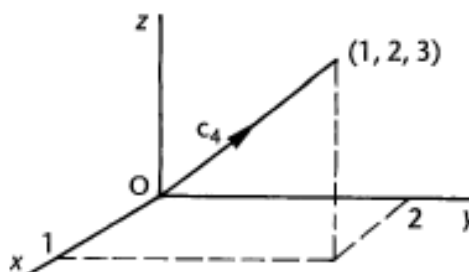
$$\int_{(0,0,0)}^{(1,2,3)} \mathbf{F} \cdot d\mathbf{r} = 0 + 0 + 27 = 27 \quad \therefore \int_{C_1+C_2+C_3} \mathbf{F} \cdot d\mathbf{r} = 27$$

- (b) If t is taken as the parameter, the parametric equations of c are

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

$$z = \dots\dots\dots$$



$$x = t; \quad y = 2t; \quad z = 3t$$

and the limits of t are

$$t = 0 \quad \text{and} \quad t = 1$$

As in Example 1, we now express everything in terms of t and complete the integral, finally getting

$$\int_{C_4} \mathbf{F} \cdot d\mathbf{r} = \dots\dots\dots$$

$$\int_{C_4} \mathbf{F} \cdot d\mathbf{r} = \frac{115}{4} = 28.75$$



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Because

$$\begin{aligned}\mathbf{F} &= 2t^3\mathbf{i} + 12t^2\mathbf{j} + 27t^3\mathbf{k} \\ d\mathbf{r} &= \mathbf{i} dx + \mathbf{j} dy + \mathbf{k} dz = \mathbf{i} dt + \mathbf{j} 2 dt + \mathbf{k} 3 dt \\ \therefore \int_{c_4} \mathbf{F} \cdot d\mathbf{r} &= \int_0^1 (2t^3\mathbf{i} + 12t^2\mathbf{j} + 27t^3\mathbf{k}) \cdot (\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}) dt \\ &= \int_0^1 (2t^3 + 24t^2 + 81t^3) dt = \int_0^1 (83t^3 + 24t^2) dt \\ &= \left[83\frac{t^4}{4} + 8t^3 \right]_0^1 = \frac{115}{4} = 28.75\end{aligned}$$

So the value of the line integral depends on the path taken between the two end points A and B

$$\begin{aligned}\text{(a)} \quad \int \mathbf{F} \cdot d\mathbf{r} \quad \text{via } c_1, c_2 \text{ and } c_3 &= 27 \\ \text{(b)} \quad \int \mathbf{F} \cdot d\mathbf{r} \quad \text{via } c_4 &= 28.75\end{aligned}$$

7.6 LINE INTEGRAL INDEPENDENT OF PATH

A line integral of the type

$$\int_C \vec{F}(\vec{r}) \cdot d\vec{r} = \int_C (F_1 dx + F_2 dy + F_3 dz) = \int_A^B \left(F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt} \right) dt$$

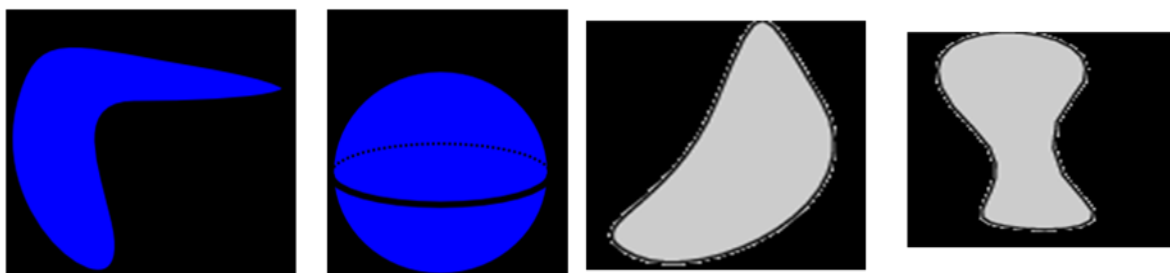
is said to be **independent of path** in a domain D in space if for every pair of end points A and B in D , in the above integral has **the same value** for all paths in D the *begin at A* and *end at B*.

7.7 LINE INTEGRAL INDEPENDENT OF PATH

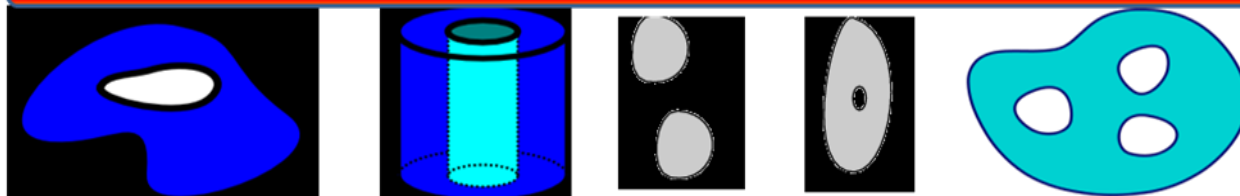
A domain D is called a ***simply connected domain*** if every closed curve in D can be continuously shrunk to any point in D without leaving D .

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SIMPLY CONNECTED DOMAINS



NOT SIMPLY CONNECTED DOMAINS

THEOREM – 3:

Let $F_1(x, y, z), F_2(x, y, z), F_3(x, y, z)$ be continuous function having continuous first partial derivatives in a domain D in space. If the line integral $\int_C (F_1 dx + F_2 dy + F_3 dz)$ is independent of path in D ,

Then $F_1 dx + F_2 dy + F_3 dz$ is exact then in D

$$\text{curl } \vec{F} = \text{curl} (F_1 \hat{i} + F_2 \hat{j} + F_3 \hat{k}) = \vec{0}$$

$$\text{i.e.} \quad \frac{\partial F_3}{\partial y} = \frac{\partial F_2}{\partial z}, \quad \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x}, \quad \frac{\partial F_2}{\partial x} = \frac{\partial F_1}{\partial y}$$

THEOREM – 4:

Let $F_1(x, y, z), F_2(x, y, z), F_3(x, y, z)$ be continuous function having continuous first partial derivatives in a domain D in space. If $\text{curl } \vec{F} = \text{curl} (F_1 \hat{i} + F_2 \hat{j} + F_3 \hat{k}) = \vec{0}$

and D is simply connected then the integral

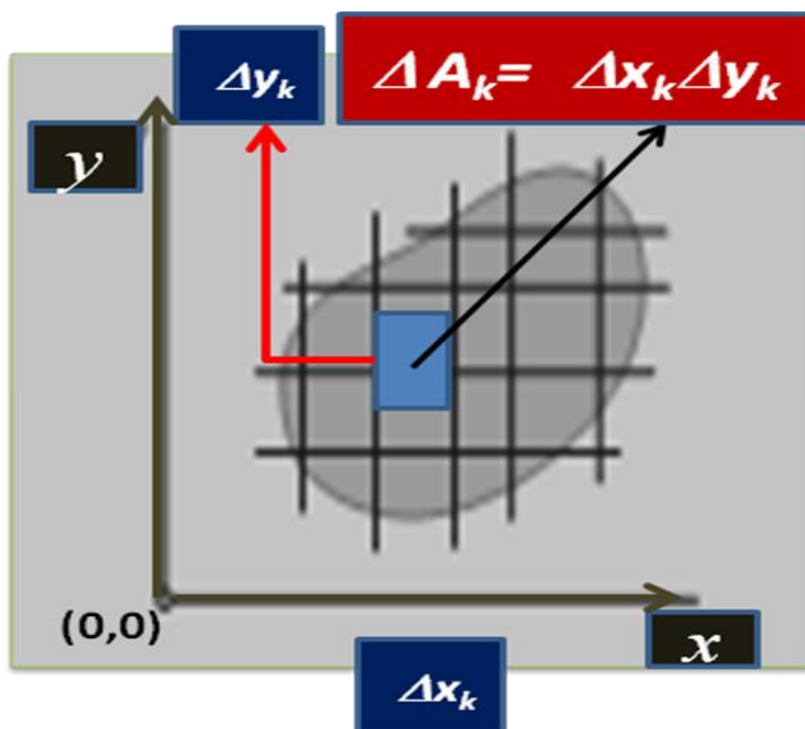
$$\int_C (F_1 dx + F_2 dy + F_3 dz) \text{ is independent of path in } D.$$

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7.8 DOUBLE INTEGRAL

Let $f(x, y)$ be any function. Let R be a region in a plane. Let us subdivide the region into a large number of small rectangular subregions. Let the k^{th} region has the dimensions say Δx_k and Δy_k .



Let (x_k, y_k) be any arbitrary point on the k^{th} subregion and $\Delta A_k = \Delta x_k \Delta y_k$ be the area of the k^{th} rectangular subregion. Next consider the sequence of real numbers $(J_1, J_2, \dots)$, where

$$J_n = \sum_{k=1}^n f(x_k, y_k) \Delta A_k$$



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Assuming that $f(x, y)$ is continuous in R and R is bounded by finitely many smooth curves. It can be shown that the sequence $(J_1, J_2, \dots)$ converges to a limit that is independent of the choice of subdivisions and corresponding points (x_k, y_k) . This limit is called the double integral of $f(x, y)$ over the region R and is denoted by

$$\iint_R f(x, y) dx dy \quad \text{or} \quad \iint_R f(x, y) dA$$

7.9 Applications of Double Integrals

- area of the region: finding area using double integral, volume of an elliptic paraboloid.
- average value of a function: calculating average storm rainfall, of a wire;
- surface integral;
- double and volume integral.

7.10 Some Properties of Double Integral

1. Properties of Linearity

$$1. \iint_A [f(x, y) + g(x, y)] dA = \iint_A f(x, y) dA + \iint_A g(x, y) dA$$

$$2. \iint_A cf(x, y) dA = c \iint_A f(x, y) dA$$

2. Comparison: If $f(x, y) \geq g(x, y)$ for all (x, y) in R , then

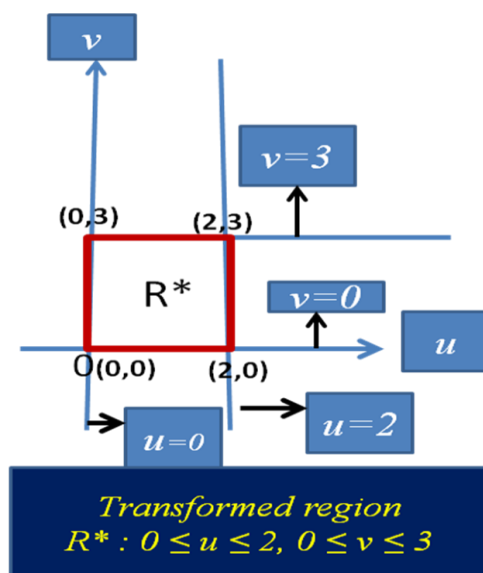
$$\iint_A f(x, y) dA \geq \iint_A g(x, y) dA$$

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Problem: Find the volume of the region of the beneath $z = 4x^2 + 9y^2$ and the rectangle with vertices $(0,0)$, $(3,0)$, $(3,2)$ and $(0,2)$.

Solution:



We know that the volume V beneath the surface $z=f(x,y)$ (>0) and above the region R in the XY plane is given by

$$V = \iint_R f(x,y) dx dy$$

$$= \int_{y=0}^2 \left(\int_{x=0}^3 (4x^2 + 9y^2) dx \right) dy = \int_{y=0}^2 \left[\frac{4x^3}{3} + 9xy^2 \right]_0^3 dy$$

$$= \int_{y=0}^2 (36 + 27y^2) dy = \left[36y + \frac{27y^3}{3} \right]_0^2 = 72 + 72 = 144$$